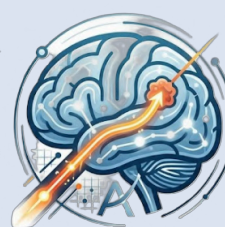




DELTA UNIVERSITY
FOR SCIENCE AND TECHNOLOGY
FACULTY OF ARTIFICIAL INTELLIGENCE



AI-Assisted 3D Brain Tumor Surgical Planning with Risk-Aware Trajectory Optimization

“Pre-Surgery Smart Planning System Using AI”

Supervisor & Members

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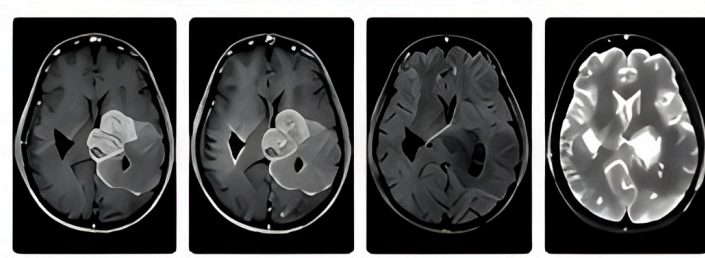
Mariam
Salah

Abstract

- AI-powered system for smart pre-surgical planning in brain tumor removal.
- Detects and segments tumor from multi-modal MRI using **SegResNet**.
- **3D visualization** of tumor with functional areas (motor, visual, language cortex).
- **A* algorithm** evaluates multiple entry points and recommends the safest, shortest surgical path avoiding critical brain regions.
- Results demonstrate **~%89** segmentation accuracy and significant reduction in surgical planning time, improving patient outcomes and procedural safety.

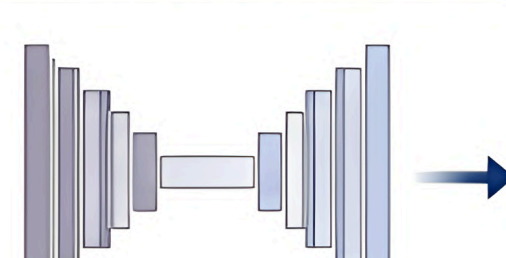
Methodology

1. INPUT MRI MODALITIES



- Volume size: $240 \times 240 \times 155$
- Cropped/Resized to: $144 \times 144 \times 144$
- Intensity normalization for better learning
- **Dataset:** BraTS 2020

2. TUMOR SEGMENTATION



Model: SegResNet
Output: 3D tumor segmentation mask

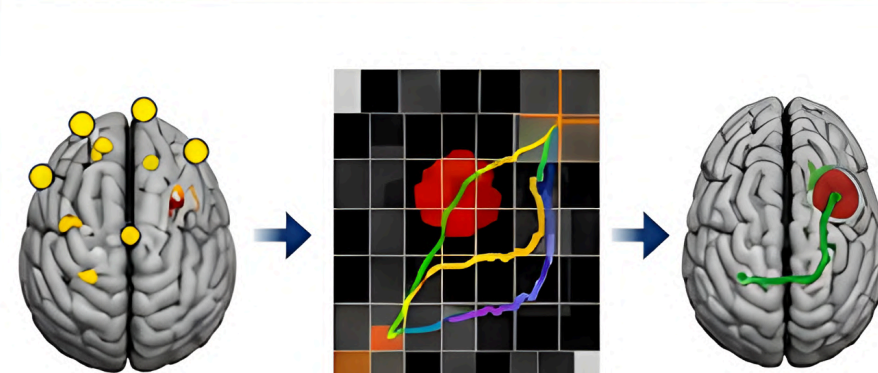


3. 3D VISUALIZATION



Legend:
Red : Tumor
Green : Motor Cortex
Blue : Language Cortex
Yellow : Visual Cortex
Interactive 3D model for surgeons

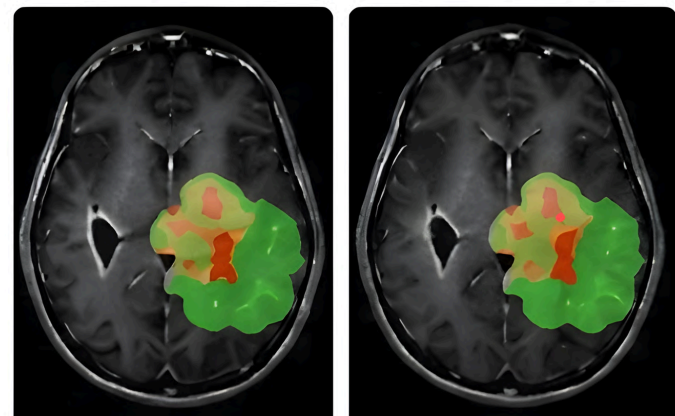
4. SURGICAL PATH PLANNING



Results

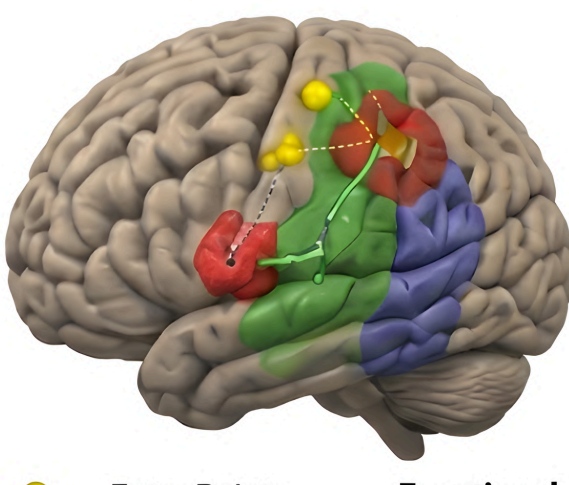
1. Brain Tumor Segmentation Results

Ground Truth Predicted (SegResNet)

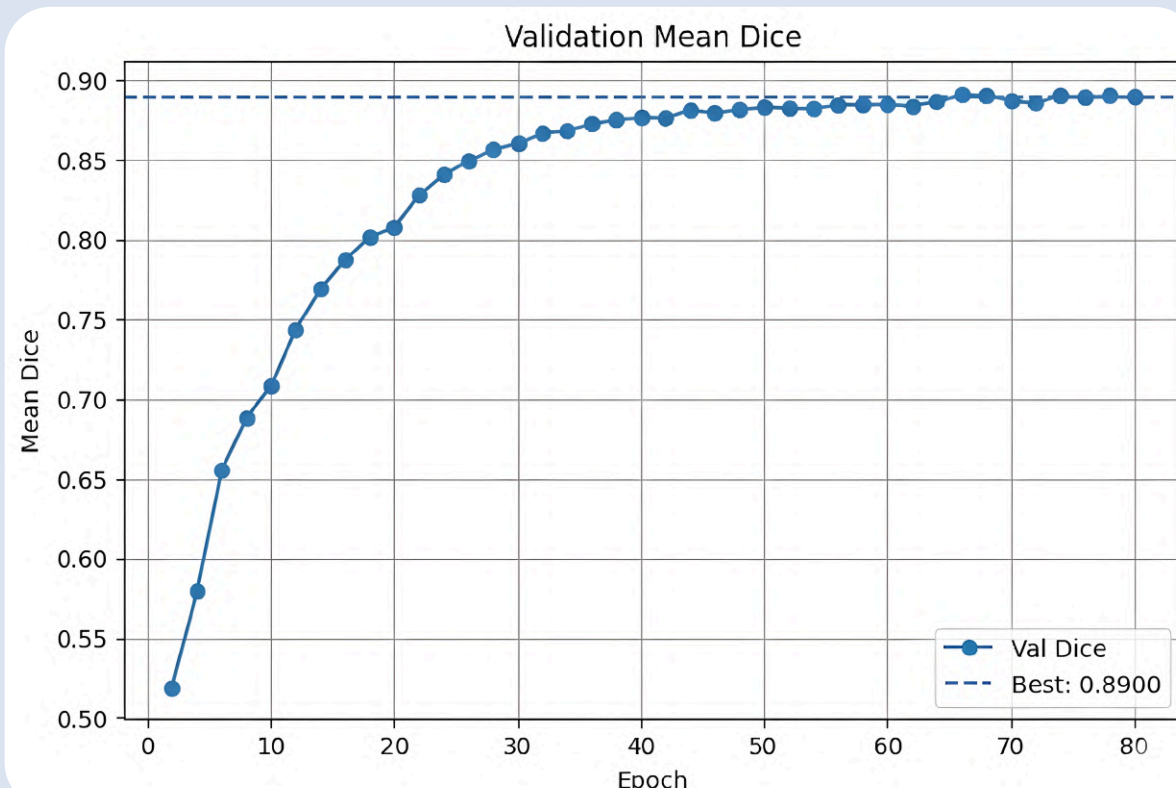


- Red : Enhancing Tumor (ET)
- Orange : Tumor Core (TC)
- Green : Whole Tumor (WT)

2. Surgical Path Planning Results



- Legend:**
Yellow : Entry Points
Green : Safest Path (A*)
Dashed line : Other Paths
Red : Tumor
- Functional Areas:**
Green : Motor Cortex
Blue : Language Cortex
Yellow : Visual Cortex



Discussion



High Segmentation Accuracy

The SegResNet model achieved ~89% Dice score, showing strong agreement with ground truth annotations.



Functional Awareness

Overlay of functional areas with tumor helps identify potential intersections and surgical risks.



Optimized Surgical Path

A* algorithm effectively finds the safest and shortest path while avoiding critical brain regions.



Clinical Relevance

Interactive 3D visualization provides intuitive insights, assisting surgeons in better decision making.



Conclusion

- Developed an end-to-end AI-based pre-surgery planning system.
- Accurate tumor detection and segmentation using SegResNet (**~89% Dice**).
- Integrated functional brain mapping for risk assessment.
- **A* based path** planning generates the safest and shortest trajectory.
- System provides interactive **3D visualization** for better surgical planning.

Future work

- Integrate **patient-specific DTI tractography** for white matter analysis.
- Incorporate real **fMRI** for more accurate functional mapping.
- Extend to **real-time surgical navigation** and intraoperative guidance.
- Improve performance using **transformer-based** models and larger datasets.
- Clinical validation through trials with **neurosurgeons**.

Acknowledgements

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GitHub

